

**Real-World Order of Operations**

Question Bank 4.07

Name \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

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*Show your work. Circle your final answer.*

**For questions 1 through 4, substitute and evaluate.**

1.  $4x^2$  when  $x = -3$

2.  $\frac{1}{2}mv^2$  when  $m = 50$  and  $v = 6$

3.  $\frac{a+b^2}{c}$  when  $a = 6$  ,  $b = 5$  ,  $c = 2$

4.  $2(x-y)^2$  when  $x = 3$  and  $y = 7$

**For questions 5 and 6, write and evaluate an expression. Give the unit.**

5. A square room of side 11 feet is floored at 3.20 dollars a square foot, with a flat 75 dollars for delivery. Find the total cost.

6. A rectangular deck 14 feet by 9 feet is stained at 1.15 dollars a square foot, with a flat 48 dollars for equipment hire. Find the total cost.

**Two formulas.**

7. Body mass index is  $B = \frac{703w}{h^2}$ , where  $w$  is a weight in pounds and  $h$  is a height in inches. Find  $B$  to the nearest tenth.

a.  $w = 150$  ,  $h = 65$

b.  $w = 180$  ,  $h = 70$

c.  $w = 200$  ,  $h = 72$

8. Kinetic energy is  $K = \frac{1}{2}mv^2$ , for mass  $m$  in kilograms and speed  $v$  in meters per second. Find  $K$  in joules.

a.  $m = 1,200$  ,  $v = 15$

b.  $m = 850$  ,  $v = 12$

c.  $m = 70$  ,  $v = 8$

**Working backwards.**

9. Find the missing value in each situation.

a.  $\frac{1}{2}mv^2 = 900$  and  $v = 6$  . Find  $m$  .

b.  $x^2 + 5 = 41$  , where  $x$  is positive. Find  $x$  .

- c. A square lawn of side  $n$  feet is turfed at 2.50 dollars a square foot with a flat 40 dollars delivery, and the total is 2,290 dollars. Find  $n$  .

**Reading the situation.**

10. Two problems use the numbers 20, 3 and 5. **A:** a gym charges 20 dollars to join plus 3 dollars for each of 5 classes. **B:** a gym charges 20 dollars for each of 3 months, plus 5 dollars to join. Find both totals, and explain why the words 'plus' and 'each' do not decide the operation.
11. A square yard of side 15 feet is turfed at 4.80 dollars a square foot, with a flat 120 dollars for delivery. Which expression gives the total cost?
- ☐ A.  $15(4.80) + 120$
  - ☐ B.  $15^2(4.80) + 120$
  - ☐ C.  $15^2(4.80 + 120)$
  - ☐ D.  $15^2 + 4.80 + 120$
12. Select ALL of the expressions whose value is 45 when  $x = 3$ .
- ☐ A.  $(5x)^2$
  - ☐ B.  $\frac{135}{x}$
  - ☐ C.  $5x^2$
  - ☐ D.  $3x^2 + 18$
  - ☐ E.  $5x \cdot 2$
  - ☐ F.  $x^2 + 36$